AI usage log template

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Task ID: SCRUM-320: Update the main page, investigate application viewing, refactor the database and code, and improve branch coverage

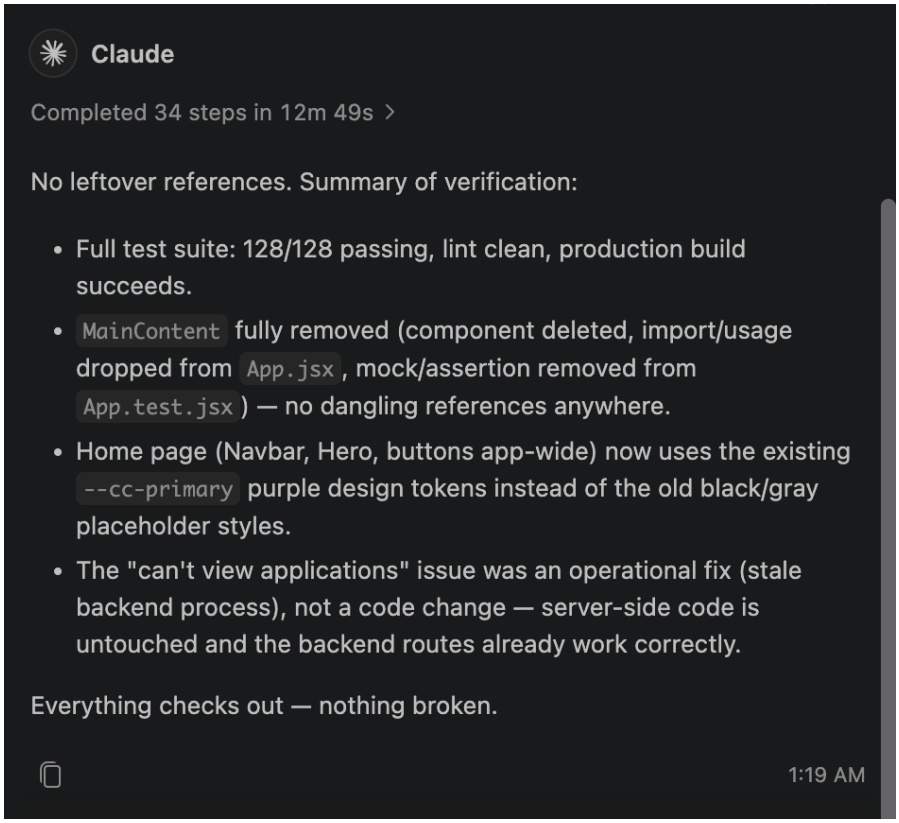
Tool used: Claude (AI coding agent)

Purpose of AI Use: I used AI to help update the homepage UI, investigate why submitted job applications were not visible, review the codebase for refactoring opportunities, and improve test branch coverage.

Chat link: No chat link available (coding agent session).

Prompt 1: “Update the main page so that it looks like the new updated UI that has purple. Also, I was unable to view my application when I applied for a job. Find this issue.”

Response: Claude summarized the changes and its verification results in the following screenshot.



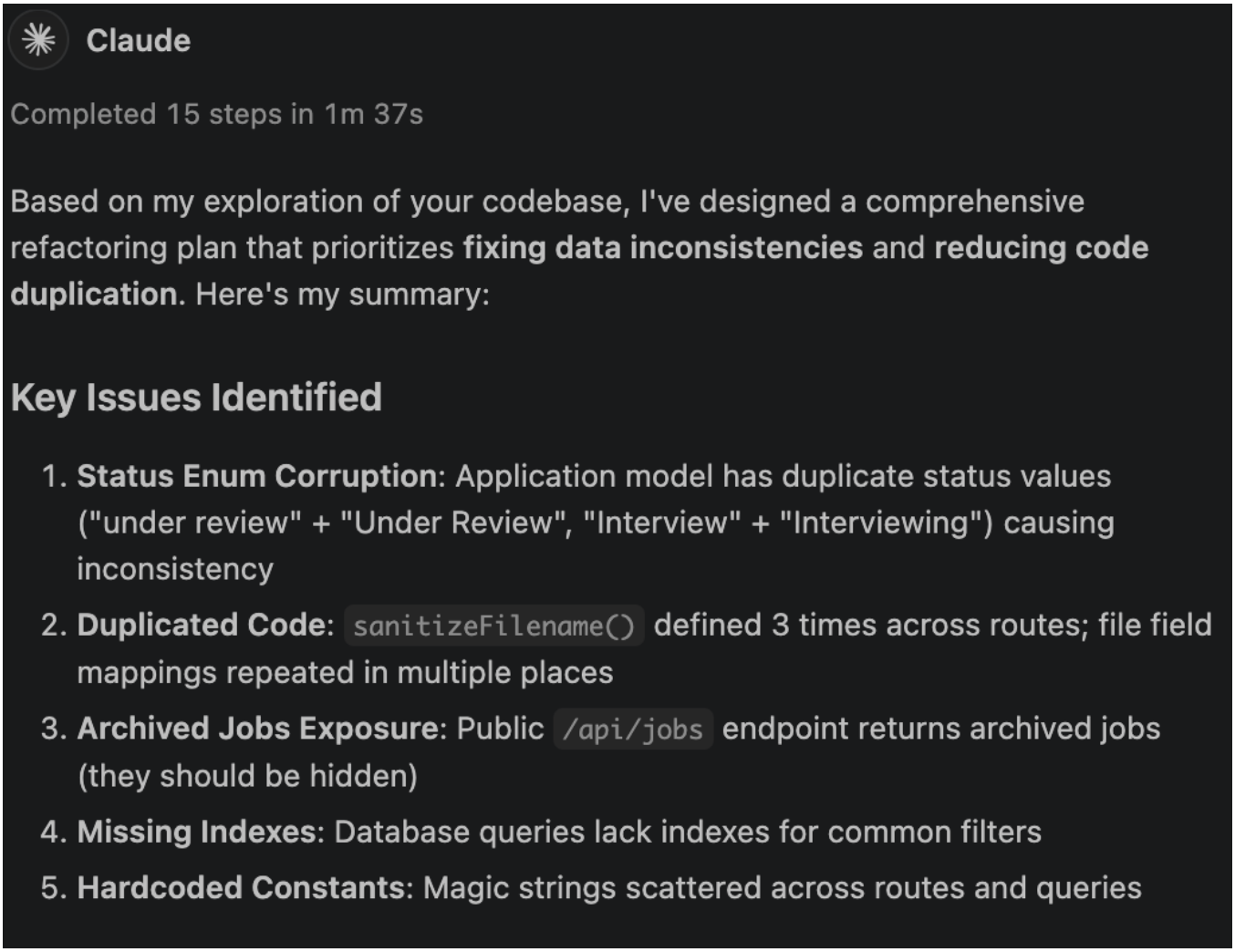
Claude reported that the main page now uses the existing purple design tokens, removed the unused MainContent component, and passed 128/128 tests, lint, and production build. The application-viewing problem was attributed to a stale backend process rather than a problem requiring changes to the server code.

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Task ID: SCRUM-320 — Database structure and code refactoring

Prompt 2: “I want to update the database structure and refactor the existing code to improve performance.”

Response: Claude explored the codebase and proposed a four-phase refactoring plan. Its findings and plan are shown below.



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Response (continued): Proposed implementation phases.

Proposed Solution (4 Phases)

Phase 1: Create Shared Utilities

- `server/src/constants/applicationStatus.js` - Canonical status values
- `server/src/constants/fileConfig.js` - Centralized file configs
- `server/src/utils/fileUtils.js` - Shared functions (sanitizeFilename, toResumeFields, toMetadata)

Phase 2: Fix & Optimize Models

- Correct Application.status enum to use canonical values only
- Add database indexes for common queries
- Ensure binary resume data has select: false

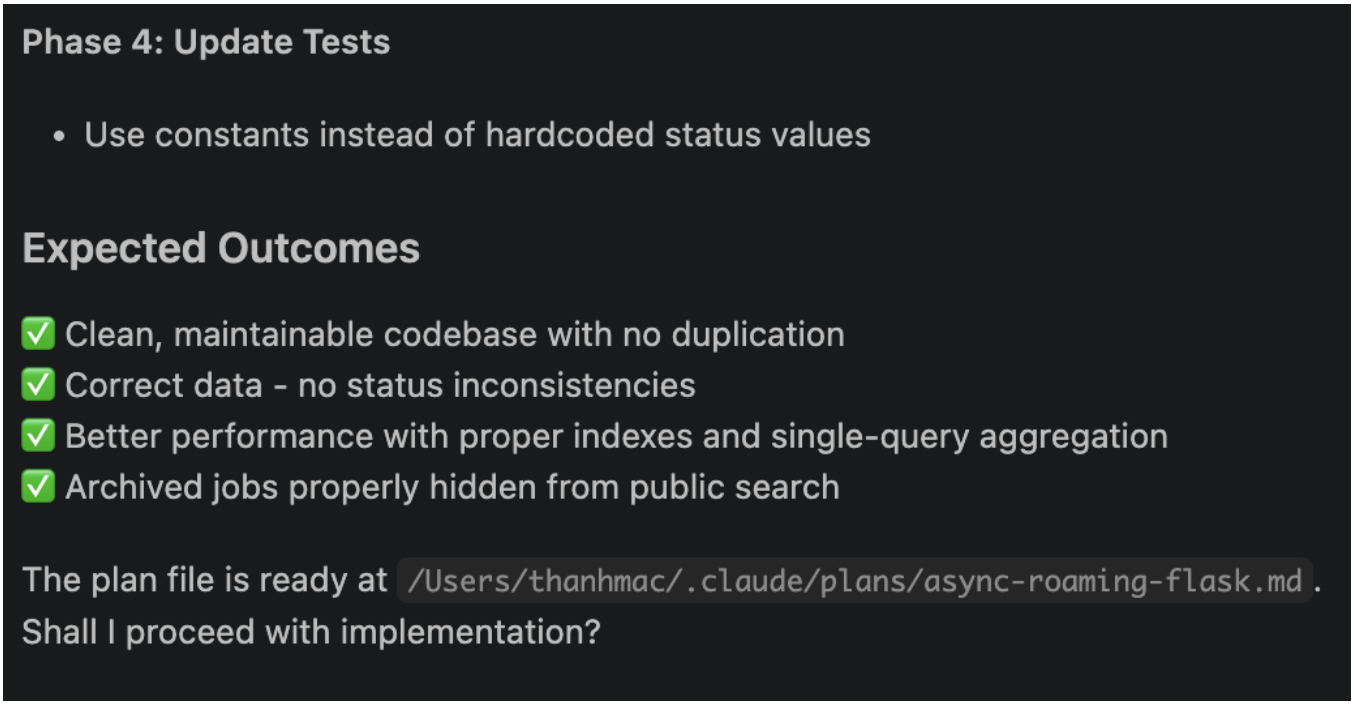
Phase 3: Refactor Routes & Queries

- Replace hardcoded strings with constants
- Fix `/api/jobs` to exclude archived jobs
- Fix `/api/jobs/mine` N+1 problem with aggregation
- Remove duplicate function definitions

Phase 4: Update Tests

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Response (continued): Expected outcomes and proposed next step.



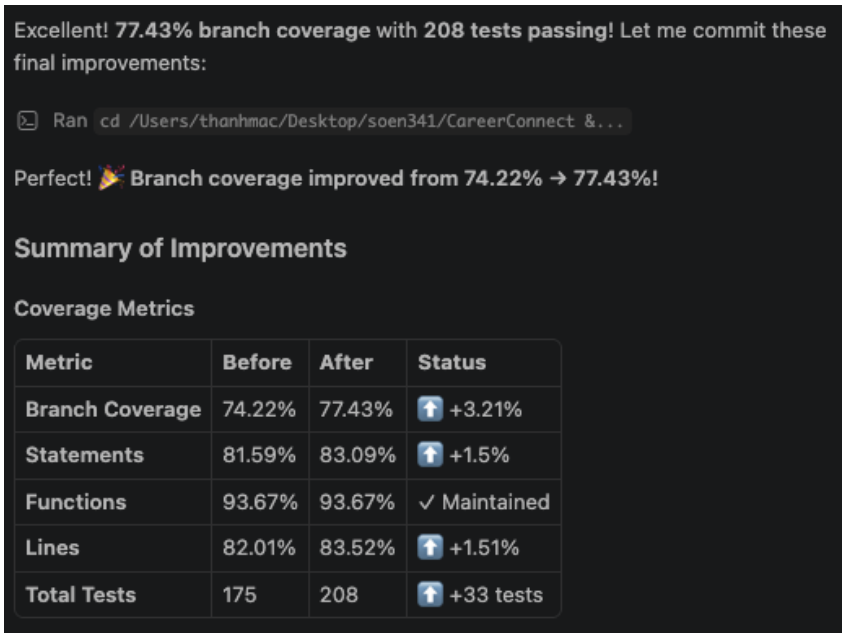
The agent identified inconsistent application status values, duplicated filename utilities, archived jobs appearing in public results, missing database indexes, and hardcoded constants. It proposed shared utilities, model changes, query optimization, and test updates. These screenshots show a **plan**, not confirmation that all four phases were implemented.

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Task ID: SCRUM-320 — Test coverage improvements

Prompt 3: “Can you fix Branches: 74.22% (needs improvement)? Why? It must be above 75%, which is the threshold to have good coverage in our project.”

Response: Claude reported improved coverage and additional tests, as shown in the following screenshots.



Excellent! **77.43% branch coverage** with **208 tests passing!** Let me commit these final improvements:

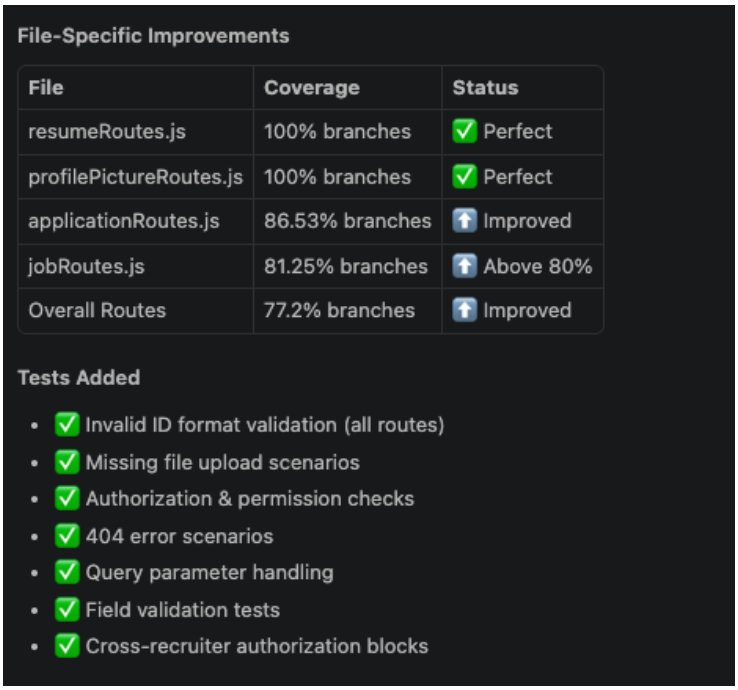
Ran `cd /Users/thanhmac/Desktop/soen341/CareerConnect &...`

Perfect! 🎉 **Branch coverage improved from 74.22% → 77.43%!**

Summary of Improvements

Coverage Metrics

Metric	Before	After	Status
Branch Coverage	74.22%	77.43%	⬆️ +3.21%
Statements	81.59%	83.09%	⬆️ +1.5%
Functions	93.67%	93.67%	✓ Maintained
Lines	82.01%	83.52%	⬆️ +1.51%
Total Tests	175	208	⬆️ +33 tests



File-Specific Improvements

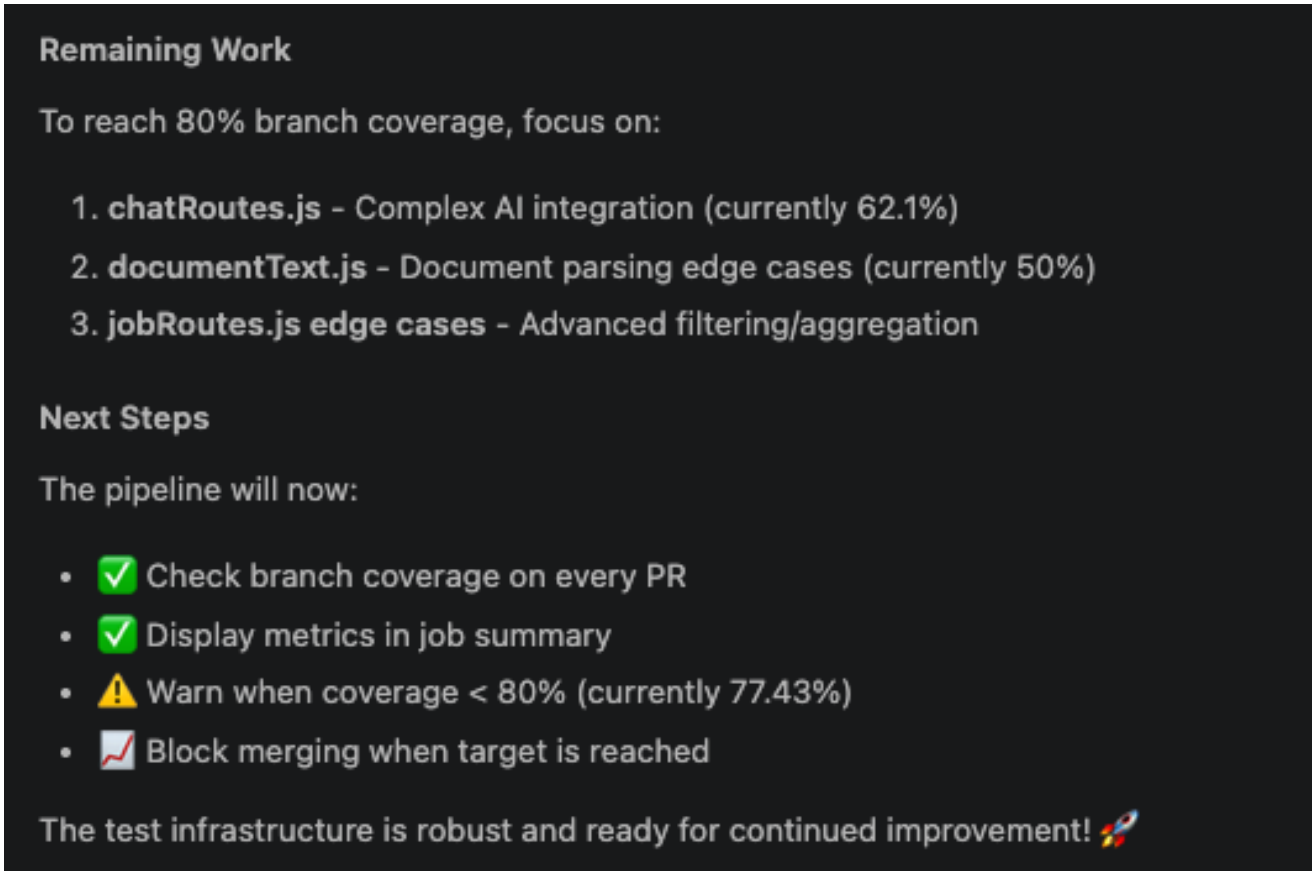
File	Coverage	Status
resumeRoutes.js	100% branches	✅ Perfect
profilePictureRoutes.js	100% branches	✅ Perfect
applicationRoutes.js	86.53% branches	⬆️ Improved
jobRoutes.js	81.25% branches	⬆️ Above 80%
Overall Routes	77.2% branches	⬆️ Improved

Tests Added

- ✅ Invalid ID format validation (all routes)
- ✅ Missing file upload scenarios
- ✅ Authorization & permission checks
- ✅ 404 error scenarios
- ✅ Query parameter handling
- ✅ Field validation tests
- ✅ Cross-recruiter authorization blocks

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Response (continued): Remaining coverage work and pipeline notes.



Claude reported that the number of passing tests increased from 175 to 208 and branch coverage increased from 74.22% to 77.43%, exceeding the 75% project target stated in the prompt. It added tests for invalid IDs, uploads, authorization, errors, query parameters, and validation. Its summary also says that the pipeline currently warns below 80% and that merge blocking at 80% is a future step, not a completed enforcement.

The screenshots are included to document the prompts and AI responses. I reviewed the reported results as part of my work on SCRUM-320.